

ASHISH MAURYA

Applied AI Engineer — LLMs, RAG, Speech AI & Agentic Systems

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EDUCATION

Bachelor of Technology, Computer Science & Engineering (AI & ML)

SRM Institute of Science and Technology, Kattankulathur, Chennai | Jun 2021 – Jun 2025

CGPA: 9.04 / 10.00

Master of Technology (M.Tech), Artificial Intelligence

Indian Institute of Technology (IIT) Patna | Jul 2026 – Jun 2028

SUMMARY

AI/ML Engineer pursuing an M.Tech in Artificial Intelligence at IIT Patna, with hands-on experience shipping production LLM, RAG, and speech-AI systems at Samespace, backed by IEEE-published research and a corresponding-author paper at ICICS 2025. Prior Research Fellow at HyperVerge Nexus and under the IndiaAI Mission grant. Core strengths: PyTorch, Transformers, and end-to-end generative AI pipelines from data to deployment.

TECHNICAL SKILLS

Languages: Python, C, C++, JavaScript, HTML, CSS

AI / Machine Learning: PyTorch, TensorFlow, Transformers, Diffusers, PEFT, Scikit-Learn

LLMs & Generative AI: RAG, FAISS, Qdrant, Weaviate, Tool Calling, Fine-Tuning, Prompt Engineering, LoRA, Stable Diffusion

Computer Vision: YOLOv8, OpenCV, Roboflow

Agentic AI: Google ADK, MCP Servers, Guardrails, Automated Evaluations

Infrastructure & Tools: Docker, Git, Linux, AWS

Databases: MySQL, PostgreSQL

EXPERIENCE

SDE-1 AI/ML Engineer

Samespace, Mumbai | Jul 2025 – Jun 2026

- Built end-to-end Retrieval-Augmented Generation pipelines using embeddings, FAISS, and hybrid retrieval for enterprise search and support workflows
- Fine-tuned foundation models for customer-specific tool-calling and instruction-following, improving task completion across production deployments
- Designed multi-agent systems on Google's ADK framework with custom MCP servers to automate
- Built multilingual TTS pipelines with code-switching and custom voice enhancements for communication products
- Implemented guardrails and automated evaluation pipelines that catch regressions pre-release, improving AI reliability
- Contributed encoder, mapper, and vocoder components to a speech-to-speech accent-translation system

AI/ML Engineer Intern

Samespace | Feb 2025 – Jul 2025

- Deployed AI solutions and integrated LLM-powered features into production systems
- Supported model serving, evaluation, and scalable AI workflow design alongside the core engineering team

Research Fellow

HyperVerge Nexus, Chennai | Feb 2024 – Jun 2024

- Developed image-generation pipelines using Stable Diffusion and LoRA fine-tuning for document and identity use cases
- Researched document forgery detection and synthetic ID generation methods
- Built generation workflows using diffusion models and the Fooocus ecosystem

Research Fellow

IndiaAI Mission | Sep 2024 – Feb 2025

- Designed and developed the ICIP framework, integrating Stable Diffusion, ControlNet, IP-Adapter, and image inpainting for controllable image generation
- Secured IndiaAI Mission research funding and presented the framework at T&F ICICS 2025

PUBLICATIONS

Obstacle Detection in Path Planning for Unmanned Aerial Vehicles Based on YOLO

IEEE | Jun 2024 | Co-Author (5 authors)

- Investigated YOLO-based object detection for autonomous UAV path planning, evaluating model efficiency and deployment constraints in resource-limited environments

ICIP: An Integrated Approach for AI Image Generation Using Stable Diffusion

T&F ICICS 2025, VVCE Mysuru | Feb 2025 | Corresponding Author

- Proposed a novel framework combining Stable Diffusion, ControlNet, IP-Adapter, and inpainting for controllable, realistic image generation; fine-tuned on custom datasets

PROJECTS

Online Video KYC System — Computer Vision, OCR, Face Verification

- Built an AI-powered KYC system for secure customer onboarding at the Standard Chartered Hackathon, combining face verification with document-validation workflows

Financial Anomaly Detection System — Machine Learning, Data Analytics

- Built a financial-transaction anomaly-detection system for the Barclays Hackathon; advanced to the final round

CERTIFICATIONS

Oracle Cloud Infrastructure 2024 Generative AI Certified Professional — Oracle, 2025